A Proof of Goldbach's Conjecture Based on the Sieve of Eratosthenes

(Author: Huajun Lv ,6014013@alu.cau.edu.cn,Yongkang City)

Abstract

This paper rigorously proves Goldbach's Conjecture utilizing the traditional Sieve of Eratosthenes. It presents the asymptotic formula and a safe lower bound for the number of ways an even number can be expressed as the sum of two odd primes, thereby proving that every even number not less than 6 can be expressed as the sum of two odd primes, and every odd number not less than 9 can be expressed as the sum of three odd primes.

Keywords: Sieve of Eratosthenes; error analysis; Goldbach conjecture; elementary number theory

1. Introduction

1.1 Overview of the Sieve of Eratosthenes

The Sieve of Eratosthenes is a classical and efficient prime sieving algorithm[1]. It obtains all primes up to a natural number N by progressively eliminating multiples of primes not exceeding $\sqrt{N}$ .The procedure is as follows: first remove multiples of 2, then multiples of 3, and continue with subsequent primes.The theoretical proportion of remaining numbers is $\prod_{i=1}^{n}\left(1-\frac{1}{p_i}\right)$.

Let $\pi(A)$ denote the number of primes not exceeding A. Then

$$\pi(A) = A \cdot \prod_{i=1}^{n}\left(1-\frac{1}{p_i}\right) + n - 1 + \delta$$

where p_i are primes $\leq \sqrt{A}$, n compensates for the primes themselves being incorrectly excluded, and the number 1 is also removed. For odd primes, the prime factor 2 is additionally excluded. The symbol δ is the error term of the sieve formula. In an asymptotic sense, the main term of the sieve systematically overestimates by approximately and no more than 12.29%[2].

2. Sieve Proof of Goldbach's Conjecture

2.1 Initial Candidate Pairs

For an even number $A \geq 6$, the number of ways to express it as $A = a + b = a + （A - a）$, with $1 \leq a \leq b \leq A - 1$, is $A/2$. These are taken as the initial candidate pairs.

2.2 Sieving out cases where a or b is divisible by p_i

For an prime $p_i \geq 2$:

· If $p_i \mid A$: remove all candidate pairs with $a \equiv 0（mod\, p_i）$ (then $b = A - a \equiv -a \equiv 0（mod\, p_i）$ automatically holds, There is no need to additionally eliminate candidate pairs satisfying $b \equiv 0（mod\, p_i）$. The proportion removed is $1/p_i$

· If $p_i \nmid A$: remove candidate pairs where $a \equiv 0（mod\, p_i）$ or $b \equiv 0（mod\, p_i）$. These two conditions are mutually exclusive (if both hold, then $A \equiv 0（mod\, p_i）$, contradiction). So the proportion removed is $2/p_i$.

After sieving by the first n primes, the remaining proportion

$$R(A) = \prod_{(p_i \mid A)\, p_i \geq 2}^{d} \left(1 - \frac{1}{p_i}\right) \cdot \prod_{(p_i \nmid A)\, p_i \geq 3}^{f=n-d} \left(1 - \frac{2}{p_i}\right) = \prod_{(p_i \mid A)\, p_i \geq 2}^{d} \frac{p_i - 1}{p_i} \cdot \prod_{(p_i \nmid A)\, p_i \geq 3}^{f} \frac{p_i - 2}{p_i}$$

2.3 Combinatorial Analysis

(1) Let $N(A)$ be the number of ways in which the even integer A can be expressed

as the sum of two prime numbers

$$N(A) = A/2 \cdot R(A) + \delta'$$

$$= A/2 \cdot \prod_{(p_i|A)p_i \geq 2}^{d} \frac{p_i - 1}{p_i} \cdot \prod_{(p_i \nmid A)p_i \geq 3}^{f} \frac{p_i - 2}{p_i} + \delta'$$

(2) Analysis of the error δ'

Part 1: The Sieve of Eratosthenes mistakenly excludes the prime p_i itself, which may need to be compensated for. The error is δ_1 ;

Part 2: The inherent error δ of the Sieve of Eratosthenes, which systematically overestimates by no more than 12.29%;

Part 3：The natural number 1 may not be excluded，the error is 0 or -1, denoted as δ_2 ;

$$\delta'=\delta_1+\delta+\delta_2 \geq \delta+\delta_2 \geq \delta - 1 \geq - A/2 \cdot R(A) * 12.29\% - 1.$$

Theorem 1 (asymptotic formula): For an even number $A \geq 6$, the number of representations as a sum of two odd primes

$$N(A) = A/2 \cdot R(A) + \delta' = A/2 \cdot \prod_{(p_i|A)p_i \geq 2}^{d} \frac{p_i - 1}{p_i} \cdot \prod_{(p_i \nmid A)p_i \geq 3}^{f} \frac{p_i - 2}{p_i} + \delta'$$

With $d + f = n$，where the error satisfies $\delta' \geq - A/2 \cdot R(A) * 12.29\% - 1$.

2.4 Lower Bound and Existence

When A has only the prime factor 2 (the minimal $R(A)$ case),

$$N(A) = A/2 \cdot 1/2 \cdot \prod_{(p_i \nmid A)p_i \geq 3}^{f} \frac{p_i - 2}{p_i} + \delta'$$

Perform an identical transformation on the product, moving the numerator $p_{i+1} - 2$ of the latter term to the denominator p_i of the former term for pairing:

$$\prod_{(p_i \nmid A) p_i \geq 3}^{f} \frac{p_i - 2}{p_i} = \frac{1}{3} \cdot \frac{3}{5} \cdot \frac{5}{7} \cdot \frac{9}{11} \cdot \ldots \cdot \frac{p_f - 2}{p_f} = \frac{1}{p_f} \prod_{(p_i \nmid A) p_i \geq 3}^{f-1} \frac{p_{i+1} - 2}{p_i}$$

Since the difference between adjacent odd primes is at least 2, each factor $(p_{i+1} - 2)/p_i \geq 1$. Therefore:

$$\prod_{(p_i \nmid A) p_i \geq 3}^{f} \frac{p_i - 2}{p_i} \geq \frac{1}{p_f} \geq \frac{1}{\sqrt{A}}$$

Hence

$$N(A) \geq A/4 \cdot \frac{1}{\sqrt{A}} + \delta' - 1 = \frac{\sqrt{A}}{4} + \delta' - 1$$

Critical Value Analysis: Because the main term systematically overestimates by no more than 12.29% in an asymptotic sense, we have:

$$N(A) \geq \frac{\sqrt{A}}{4} \text{（} 1 - 12.29\% \text{）} - 1$$

To ensure $N(A) \geq 1$, we require $A \geq 84$.

Theorem 2 (Goldbach conjecture, even case): For every even number $A \geq 6$, there exist odd primes p, q such that $A = p + q$.

Proof: For $A \geq 84$, the lower bound gives $N(A) \geq 1$, so at least one representation exists. For $6 \leq A \leq 84$, direct verification

$\cdot 6 = 3 + 3$

$\cdot 8 = 3 + 5$

$\cdot 10 = 3 + 7 = 5 + 5$

$\cdot 12 = 5 + 7$

……

$\cdot 84 = 5 + 79$

confirms the statement.

Theorem 3 (Goldbach conjecture, odd case): For every odd number $B \geq 9$, there exist odd primes p_1, p_2, p_3 such that $B = p_1 + p_2 + p_3$.

Proof: Choose an odd prime $p_1 \geq 3$ such that $B - p_1$ is even and ≥ 6 (e.g., $p_1 = 3$ works because $B - 3$ is even and ≥ 6). By Theorem 3, $B - p_1 = p_2 + p_3$ for some odd primes p_2, p_3. Hence $B = p_1 + p_2 + p_3$.

3. Conclusion

Sieve proof of the Goldbach conjecture: Using the optimized sieve method, we obtain an asymptotic formula for the number of representations of an even number as a sum of two odd primes:

$$N(A) = A/2 \cdot R(A) + \delta' = A/2 \cdot \prod_{(p_i \mid A) p_i \geq 2}^{d} \frac{p_i - 1}{p_i} \cdot \prod_{(p_i \nmid A) p_i \geq 3}^{f} \frac{p_i - 2}{p_i} + \delta'$$

With $d + f = n$, $\delta' \geq - A/2 \cdot R(A) * 12.29\% - 1$.

A lower bound analysis shows $N(A) \geq 1$ for all even $A \geq 84$, and direct verification covers smaller cases. Consequently, the Goldbach conjecture holds for all even numbers ≥ 6 and odd numbers ≥ 9.

References

[1] Chen J R, Wang T Z. On the Goldbach problem[J]. Acta Mathematica Sinica, 1989, 32(5): 702-718.

[2] Cojocaru, A. C., & Murty, M. R. (2005). An Introduction to Sieve Methods and Their Applications. Cambridge University Press.